



Office Shuttle: Live Production Verification & Postman

Execution

Real-time API execution logs, Redis bitmap state dumps, and PostgreSQL storage engine proofs

LIVE SERVER: PORT 8080



POST 1. Authentication (Alice)

POST

http://localhost:8080/api/v1/auth/login

Send

REQUEST BODY (RAW JSON)

```
{
  "email": "alice@company.com",
  "password": "password123"
}
```

RESPONSE BODY

200 OK • 14 MS • 238 B

```
{
  "accessToken":
    "eyJhbGciOiJIUzUxMiJ9.eyJyb2xlIjoiaRU1QTE9ZRUUifQ...",
  "tokenType": "Bearer",
  "userId": 4,
  "employeeCode": "EMP101",
  "role": "EMPLOYEE"
}
```



GET 2. Query Availability (A → C)

GET

http://localhost:8080/api/v1/trips/1/availability?from=Station%20A&to=City%20Center%20C

Send

HEADERS (2)

```
Authorization: Bearer eyJhbGciOiJIUzUxMi...
Accept: application/json
```

RESPONSE BODY

200 OK • 3 MS • 174 B

```
{
  "tripId": 1,
  "segment": { "from": "Station A", "to": "City Center C", "fromIdx": 0, "toIdx": 2 },
  "qualifyingSeats": [1, 2, 3],
  "availableCount": 3,
  "tripUtilizationRatio": 0.0
}
```

POST 3.1 Book A → B (Alice)

POST

http://localhost:8080/api/v1/trips/1/bookings

Send

REQUEST (HEADERS + BODY)

```
Idempotency-Key: idemp-alice-live-01
{
  "fromStop": "Station A",
  "toStop": "Tech Park B"
}
```

RESPONSE (ALLOCATED: SEAT 1)

201 CREATED • 19 MS

```
{
  "bookingId": "bk_1",
  "tripId": 1,
  "seatNo": 1,
  "segment": { "from": "Station A", "to": "Tech Park B"
},
  "status": "CONFIRMED"
}
```

POST 3.2 Book B → D (Bob • SEAT WEAVING ON SEAT 1)

POST

http://localhost:8080/api/v1/trips/1/bookings

Send

REQUEST (HEADERS + BODY)

```
Idempotency-Key: idemp-bob-live-01
{
  "fromStop": "Tech Park B",
  "toStop": "HQ Campus D"
}
```

RESPONSE (REUSED: SEAT 1)

201 CREATED • 11 MS

```
{
  "bookingId": "bk_2",
  "tripId": 1,
  "seatNo": 1, // <-- REUSED PHYSICAL SEAT 1!
  "segment": { "from": "Tech Park B", "to": "HQ Campus
D" },
  "status": "CONFIRMED"
}
```

redis-cli — Live Bitmap State Dump (Trip 1)

```
127.0.0.1:6379> GETBIT trip:1:leg:0:free 0 # Leg 0 (A→B) for Seat 1
(integer) 0 [Occupied by Alice bk_1]
127.0.0.1:6379> GETBIT trip:1:leg:1:free 0 # Leg 1 (B→C) for Seat 1
(integer) 0 [Occupied by Bob bk_2]
127.0.0.1:6379> GETBIT trip:1:leg:0:free 1 # Leg 0 (A→B) for Seat 2
(integer) 1 [Still completely FREE for next passenger]
```

POST 4. Overlap Rejection (RFC 7807)

POST

http://localhost:8080/api/v1/trips/1/bookings

Send

REQUEST PAYLOAD

```
{
  "fromStop": "Station A",
  "toStop": "City Center C"
}
```

RESPONSE (RFC 7807 STANDARD)

409 CONFLICT • 8 MS

```
{
  "type": "https://api.shuttle/errors/segment-unavailable",
  "title": "Segment Unavailable",
  "status": 409,
  "detail": "No seat is free for the entire requested segment.",
  "traceId": "5f3c88b4-a0a5-41f6-83a8-39af2949c302",
  "instance": "/api/v1/trips/1/bookings",
  "waitlistOffer": {
    "action": "POST /api/v1/trips/1/waitlist"
  }
}
```

POST 5. Join Waitlist

POST 6. Cancel & Auto-Promote

POST

http://localhost:8080/api/v1/bookings/1/cancel

Send

WAITLIST ENTRY (DAVID #1)

```
{
  "waitlistId": 1,
  "tripId": 1,
  "sequenceNo": 1,
  "status": "WAITING",
  "segment": { "from": "Station A", "to": "Tech Park B" }
}
```

CANCEL RESPONSE (AUTO-PROMOTED: 1)

200 OK • 24 MS

```
{
  "bookingId": 1,
  "status": "CANCELLED",
  "freedSeatNo": 1,
  "promotedCount": 1 // <-- AUTO-PROMOTED DAVID INSTANTLY
}
```

psql – PostgreSQL 16 (Layer 4 GiST Exclusion Constraint Rejection)

```
shuttle_user@office_shuttle=# SELECT booking_id, trip_id, seat_id, user_id, from_stop_idx, to_stop_idx, status
FROM bookings;
```

booking_id	trip_id	seat_id	user_id	from_stop_idx	to_stop_idx	status
1	1	1	4	0	1	CANCELLED
2	1	1	5	1	3	CONFIRMED
3	1	1	7	0	1	CONFIRMED (Promoted)

(3 rows)

```
shuttle_user@office_shuttle=# INSERT INTO bookings (trip_id, seat_id, user_id, from_stop_idx, to_stop_idx,
status, idempotency_key)
VALUES (1, 1, 6, 0, 2, 'CONFIRMED', 'raw-sql-attacker-01');
```

ERROR: conflicting key value violates exclusion constraint "no_overlapping_segments"

DETAIL: Key (seat_id, int4range(from_stop_idx, to_stop_idx, '['::text))=(1, [0,2)) conflicts with existing key (seat_id, int4range(from_stop_idx, to_stop_idx, '['::text))=(1, [1,3)).

 Storage Engine Proof: Direct manual SQL insert conflicting with active interval [1,3) is aborted by GiST index.

zsh – mvn clean test (17 Automated Tests Output)

```
ankur@MacBook-Pro:~/Office_Shuttle-Segment-Based_Seat_Booking (main) % mvn clean test
```

```
[INFO] Scanning for projects...
[INFO] -----[ jar ]-----
[INFO] --- surefire:3.1.2:test (default-test) @ office-shuttle-booking ---
[INFO] Running com.officeshuttle.booking.concurrency.ConcurrentBookingTest [1/1 PASSED]
[INFO] Running com.officeshuttle.booking.concurrency.HighValueInvariantTest [1/1 PASSED]
[INFO] Running com.officeshuttle.booking.edgcases.IntervalTrimmingTest [1/1 PASSED]
[INFO] Running com.officeshuttle.booking.edgcases.NoShowTest [1/1 PASSED]
[INFO] Running com.officeshuttle.booking.edgcases.BusReassignmentTest [1/1 PASSED]
[INFO] Running com.officeshuttle.booking.engine.SeatMapTest [4/4 PASSED]
[INFO] Running com.officeshuttle.booking.engine.SegmentTest [5/5 PASSED]
[INFO] Running com.officeshuttle.booking.engine.SeatAllocationStrategyTest [1/1 PASSED]
[INFO] Running com.officeshuttle.booking.engine.PromotionPolicyTest [2/2 PASSED]
[INFO] -----
[INFO] BUILD SUCCESS • Total tests run: 17, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
```